

Machine Learning Application for Prediction of EV Charging Demand for the Scenario of Agartala, India

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Abstract—Electric vehicle (EV) is approaching to the market in huge scale. Futuristic planning is necessary to accommodate the EVs in changing station as well as in electrical system. Therefore, the possible charging demand by the EVs should be predicted for proper capable infrastructure. Machine Learning (ML) is the best technique for prediction. This current paper presents a comparative study among three ML techniques to determine the ideal ML technique to predict required SOC of EVs. Existing vehicle flow of Agartala city is used to train the ML model. The results show that, among CG-SVM, bagged tree regressor, and boosted tree regressor; CG-SVM outperforms the other two ML techniques with respect to performance metrics, such as- MSE, MAE, R-squared value and RMSE also been observed. CG-SVM has lesser training time and higher prediction speed. Thereby, establishing CG-SVM as suitable ML technique to predict the required SOC of EVs.

Keywords—Electric vehicle, machine learning, prediction, state of charge.

I. INTRODUCTION

Electric vehicle (EV) is the sustainable transportation for future. EV has become necessary due to environmental worries and diminution of fossil oil. EV are developing more attractiveness for zero emission and low running cost compared to oil based vehicle. EV charging infrastructure is required to charge the EVs. Increasing number of EVs create additional effect on power system network. Also, the planning of EV charging station (EVCS) is really a difficult job, because of the appropriate estimation of load due to EVs. Huge number of EV penetration makes significantly surge the power demand and peak load, especially in large adoption areas like major cities. Without appropriate infrastructure of EVCS and distribution network with huge power demand due to EVs may cause burden on electrical system and low power quality. As a result, a precise prediction of energy demand for EV charging is mandatory for proper planning of the electrical infrastructures.

Estimating the state of charge (SOC) requirement of the EVs is newly took attention by the researchers of power system, transportation system and vehicular technology, because of

rising number of EVs. An algorithm has been developed in [1] based on machine learning to predict the behaviour of EV users. SOC of general li-ion battery is estimated using GPR and linear regression in [2]. Various machine learning techniques have been applied and real data have been used to test the robustness of the model. In [3], charging demand of EVs has been predicted for a urban area. Closed-circuit televisions (CCTVs) have been used to collect the traffic data from the roads. In [4], the EV load on power system load profile has been designed by the prediction of EV charging demand with a proposed method. In [5], EV load is forecasted to make efficient strategy for EV charging. Three case studies have been evaluated, i.e. uncontrolled charging, controlled charging and smart charging. EV range prediction has been performed in [6] by considering different internal and external influencing parameters. Also, present problem and possible solutions have been discussed. EV load has been estimated in [7], in 1 hour time interval for 24 hours. Different prediction techniques have been applied and a system has been designed for mobile application. Autoregressive integrated moving average (ARIMA) has been employed for prediction of EV charging demand in [8], [9]. Modified pattern-based sequence forecasting (MPSF) has been compared with KNN, random forest and support vector machine for prediction the real EV energy requirement in EVCSs. These have been verified for UK based scenarios. EV SOC has been forecasted in [10] for various speed scenarios. Federated learning based energy requirement estimation has been done in [11]. Authors in [12], have shown EV charging discharging strategy based on EV required SOC prediction. In [13], exponential smoothing techniques has been used to predict EV charging demand. In [14], data have been collected from conventional fossil fuel base vehicle for a urban area for prediction. However, the real data related to EVs and traffic flow are very limited. Moreover, only few literature are available on prediction of EV SOC requirement for future planning of EVCS in a practical city.

In this paper, the SOC requirement is predicted using CG-SVM, bagged tree regressor, and boosted tree regressor techniques and a comparative study is shown to check the

performance. For the case study, Agartala city is chosen, which is a capital of Tripura state in India. A number of EVs is also coming to Agartala and one EVCS is already installed. Gradually, fossil fuel based vehicle will be replaced by EVs. However, the vehicle trip or users behaviour will remain same. As, at this present, EVs are not available in Agartala, therefore, the traffic flow of conventional vehicles are taken. Using the existing data related to flow of the conventional vehicles, the SOC requirement of future EVs is predicted in this work. The authors have surveyed on different types vehicle, such as, two wheeler, three wheeler, private car, public taxi and commercial goods vehicle. Their daily trip millage, first trip distance, trip start location, trip end location, trip start time and trip end time. Same behaviours are taken for the future EVs and SOC requirements are predicted using machine learning. Machine learning is process of training the machine based upon historical data to predict the future course of action or predict future values. Machine learning are now being adopted throughout the world to different engineering problems. Hence, machine learning can be an ideal solution to predict the required SOC of EVs.

II. METHODOLOGY

$SOC_{requirement}$ – Required SOC by an EV.

$SOC_{departure}$ – Departure SOC of the EV.

$SOC_{arrival}$ – Arrival SOC of the EV.

$Trip_{subsequence}$ – Travel distance after charging station.

$Trip_{first}$ – Travel distance before charging station.

AER – All-electric range.

The SOC requirement of an EV depends on arrival SOC and departure SOC, can be calculated as [15]:

$$SOC_{requirement} = SOC_{departure} - SOC_{arrival} \quad (1)$$

where departure SOC depends on the estimated trip distance after attending charging station and all-electric range of the EV. Similarly arrival SOC is calculated using trip distance before coming to charging station and AER.

$$SOC_{departure} = \frac{Trip_{subsequence}}{AER} \quad (2)$$

$$SOC_{arrival} = \frac{Trip_{first}}{AER} \quad (3)$$

The paper aims to predict the SOC requirement of EVs using different ML algorithms. The objective of the study is to determine the suitable ML algorithm for SOC prediction. For the case study, Agartala, India has been chosen as the place of study. As, EVs are very less in numbers in the city, survey data of conventional vehicles are taken into consideration. Total sample size of 2000 conventional vehicles are noted and subsequently their behaviour is converted into equivalent EV behaviour. Arrival time, departure time, arrival SOC, departure SOC, required time for charging, and required SOC are noted for these vehicles. After that, the data is pre-processed and subjected to ML algorithms for prediction. Figure 1 presents proposed methodology.

For the present study, regression learner app in MATLAB/Simulink environment is used. To protect against overfitting of the predicted result, 10-fold cross validation has been used. The trained dataset is then subjected to test using unseen data values to evaluate the efficacy. For the present study, coarse gaussian support vector machine (CG-SVM), bagged tree regressor, and boosted tree regressor has been utilized. The configuration of the system is 16 GB DDR4 RAM, 10th generation Intel core-i7 processor, NVIDIA GeForce GTX 1650 Ti 4GB graphic processor, and 1 TB SSD.

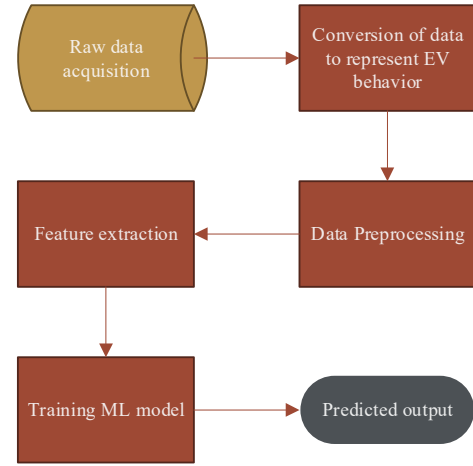


Fig. 1. Flowchart of the proposed methodology

III. RESULTS AND DISCUSSIONS

The present section summarizes and analyses the results obtained to evaluate the efficacy of tested ML algorithms. The obtained results indicate that, proposed ML techniques can efficiently predict the required SOC of EVs.

To evaluate the performance, root mean square error (RMSE), R squared error, mean squared error (MSE), mean absolute error (MAE) are taken as performance metrics to evaluate the performance of the implemented techniques. T

MSE is the squared value of difference between original and predicted values divided by total number of observations. MAE is the difference between original and predicted values, divided by the total number of observations. RMSE is the root mean squared value of MSE, and R- squared value denotes the how the predicted values fit in comparison to the original values. The value of R-squared value ranges in between 0 to 1. Where 1 represents the perfect performance of the model. Hence, it is evident that the lower values of MSE, MAE, and RMSE indicates superior performance of the ML model. The lower values of RMSE, MSE, and MAE indicates the efficient performance of ML algorithms. R-squared values closer to 1 also endorses the superiority of the ML algorithms. The following equations can be used to determine the performance of the said tested techniques:

$$SMSE = \frac{1}{n} \sum_{i=1}^n (Y_i - \hat{Y}_i)^2 \quad (4)$$

$$MAE = \frac{1}{n} \sum_{i=1}^n |Y_i - \hat{Y}_i| \quad (5)$$

$$RMSE = \sqrt{MSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (Y_i - \hat{Y}_i)^2} \quad (6)$$

$$R^2 = \frac{\sum_{i=1}^n (Y_i - Y'_i)(\hat{Y}_i - \hat{Y}'_i)}{\sqrt{\sum_{i=1}^n (Y_i - Y'_i)^2 \sum_{i=1}^n (\hat{Y}_i - \hat{Y}'_i)^2}} \quad (7)$$

Here, Y_i is the actual output, $\hat{Y}_i$ is the predicted output, Y'_i is the average of the predicted output, Y'_i is the average of the actual output, and n is the number of samples.

Table I presents the performance of the investigated ML algorithms. It is evident from the obtained results that, CG-SVM outperforms all the other techniques. It is evident that, the CG-SVM outperforms all the other models w.r.t all the tested performance parameters.

TABLE I. PERFORMANCE OF THE INVESTIGATED ML ALGORITHMS

ML techniques	RMSE	R squared	MSE	MAE
CG- SVM	0.6144	1	0.3775	0.43887
Bagged tree	1.8111	0.99	3.28	1.2253
Boosted tree	1.8929	0.98	3.5831	1.5988

Figure 2, 3 and 4 represent the response plot of three ML techniques. The response plot demonstrates the fitting of original points and predicted datapoints. The figures indicate that the all the three models perform commendably to fit the predicted datapoints. Moreover, it is also seen that the predicted datapoints are not following any patterns, rather these are randomly placed. This ensures the models are not fitted poorly. Among the 3 trained models, CG-SVM is best fitted model and can be selected as the ideal model to predict the required SOC.

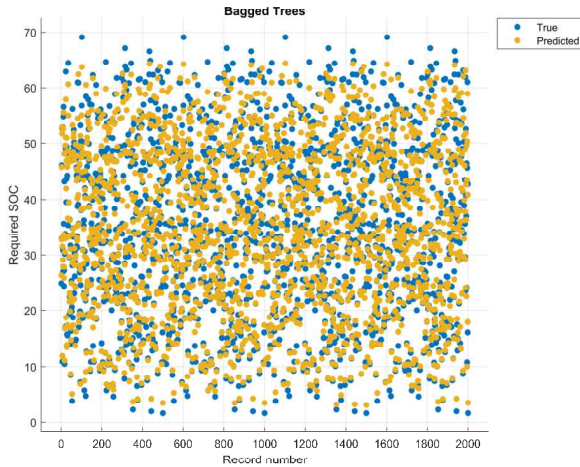


Fig. 2. Response plot of Bagged Trees

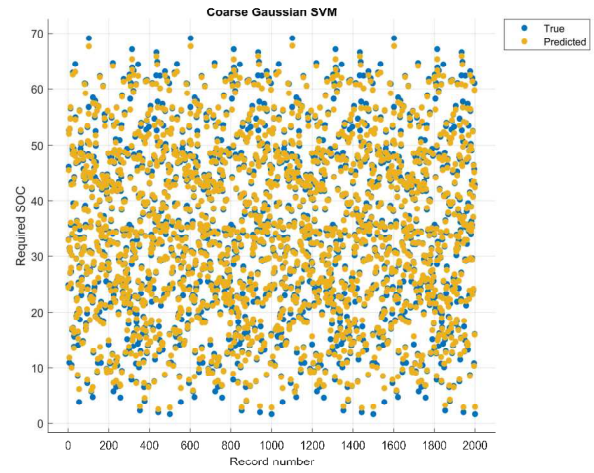


Fig. 3. Response plot of CG-SVM

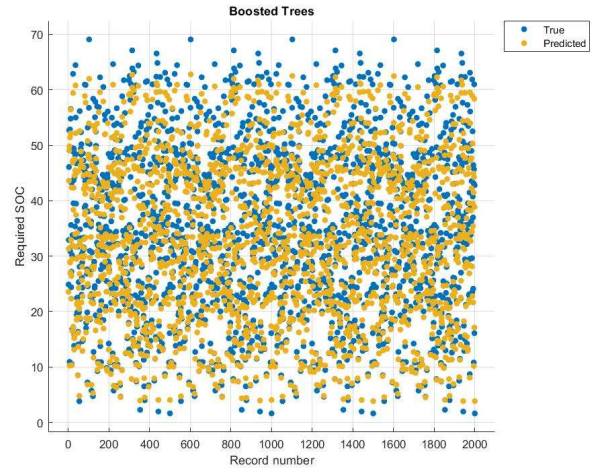


Fig. 4. Response plot of Boosted Trees

Figure 5, 6 and 7 show the residuals plot. The residual plot demonstrates the distance between predicted and actual values. The present study plots the predicted responses against the true responses. The obtained results clearly indicates that, all the three models can efficiently predict the required SOC. Among them, it is again evident that, CG-SVM is the most efficient ML model which can effectively predict the required SOC.

The figure 8, 9 and 10 demonstrate the plot of predicted response against the true response. Perfectly predicted model will have values same as true response. Hence, through examining these figures, it can be determined that, the models can accurately predict the required SOC. However, in this case also, CG-SVM is found to be superior to the other two regression models.

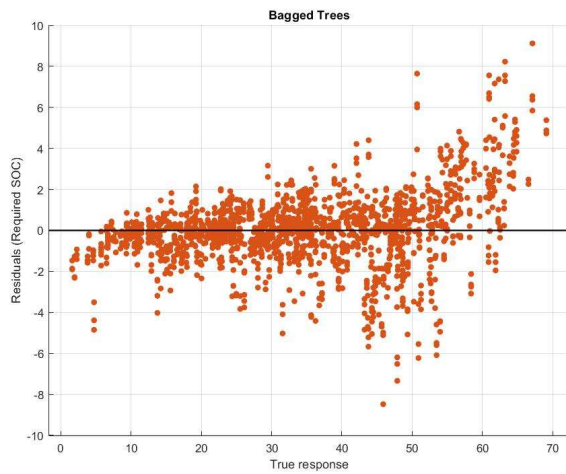


Fig. 5. Residuals plot of Bagged Trees

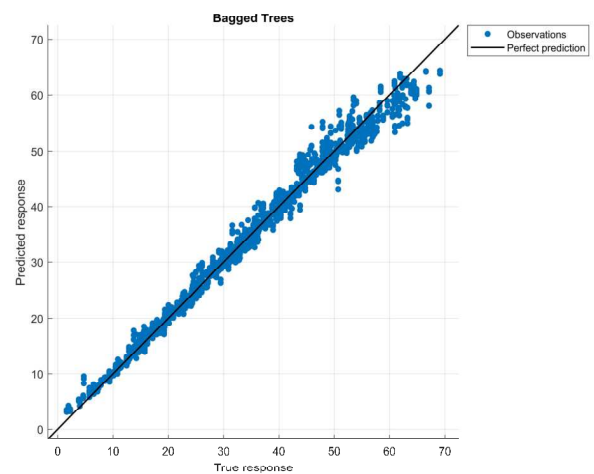


Fig. 8. Predicted response against the true response of Bagged Trees

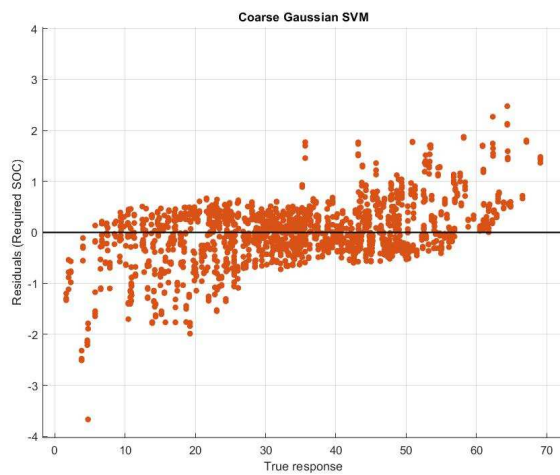


Fig. 6. Residuals plot of CG-SVM

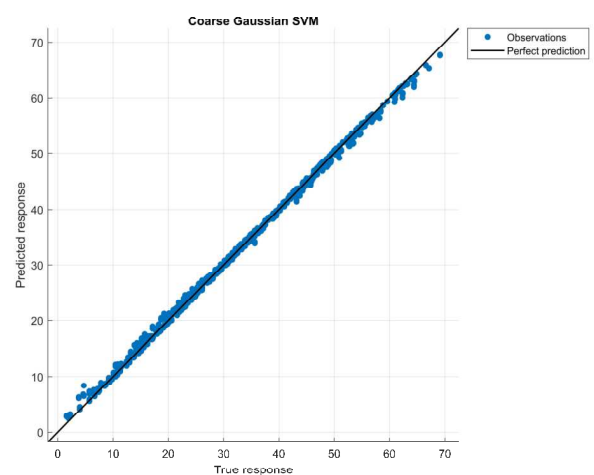


Fig. 9. Predicted response against the true response of CG-SVM

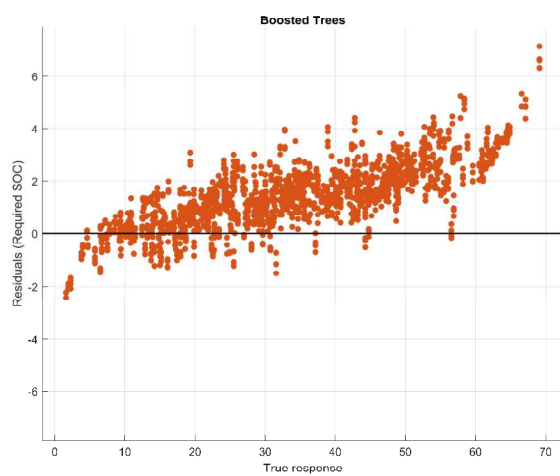


Fig. 7. Residuals plot of Boosted Trees

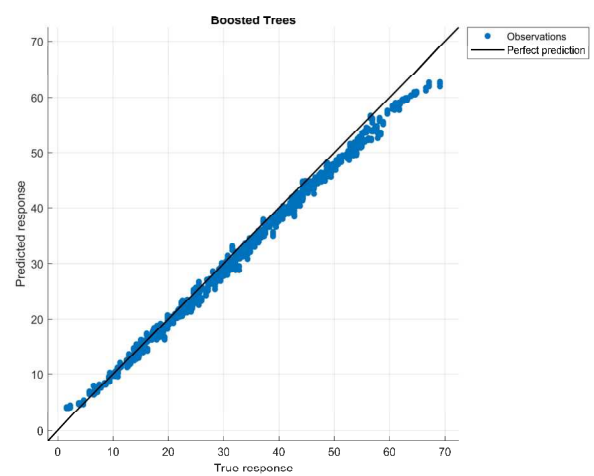


Fig. 10. Predicted response against the true response of Boosted Trees

Hence, analyzing all the plots and table, it can be concluded that, CG-SVM is the preferred ML model for prediction of required SOC of EVs.

TABLE II. TRAINING TIME AND OBSERVATION TIME OF ML ALGORITHMS

ML techniques	Training speed (s)	Prediction speed (obs/sec)
CG- SVM	7.244	~32000
Bagged tree	8.951	~9500
Boosted tree	9.623	~14000

The training time and observation time is an important parameter for determining suitable ML model. It can be observed in Table II, the training time of CG-SVM is comparatively less than the other two tested ML models. CG-SVM also has the highest observation speed among all three tested algorithms.

IV. CONCLUSION

For proper infrastructure of EVCS and distribution network, a precise prediction of SOC demand for EV charging is mandatory. However, real data related to EVs is still not available for all the location. Therefore, in this paper, the traffic flow of existing vehicles at Agartala has been taken as input data. Machine learning is the most powerful tool for prediction, which is used to predict the SOC requirement of the EVs. A comparative study of three models has been carried out. Analysing all the results, it can be concluded that, CG-SVM is the most suitable ML model among all three tested algorithms. CG-SVM outperforms the other two tested algorithms. RMSE, MAE, R squared values, and MSE values endorse the efficacy of CG-SVM to predict the required SOC of EVs. In addition, CG-SVM, also possesses the lowest training time. Hence, CG-SVM is the most suitable ML algorithm as it has superior performance and lesser computation time.

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